# Homework 2.8 #1

Student Name: ______________________________

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **10**

Instructor Name: **Mamani Velasco, Ruben**

## Question 1 of 10

Suppose that the functions $g$ and $h$ are defined for all real numbers $x$ as follows.

$$g(x) = x - 3$$
$$h(x) = 2x^2$$

Find the following.

$(g \cdot h)(x) =$ ______________________________

$(g + h)(x) =$ ______________________________

$(g - h)(-1) =$ ________

## Question 2 of 10

Suppose that the functions $f$ and $g$ are defined as follows.

$$f(x) = 2x^2 - 7$$
$$g(x) = -7x + 6$$

(a) Find $\left(\frac{f}{g}\right)(-1)$.

(b) Find all values that are NOT in the domain of $\frac{f}{g}$.

If there is more than one value, separate them with commas.

## Question 3 of 10

The functions $r$ and $s$ are defined as follows.

$$r(x) = x + 2$$
$$s(x) = -x^2 + 1$$

Find the value of $s(r(-5))$.

$s(r(-5)) = \square$

## Question 4 of 10

Suppose that the functions $p$ and $q$ are defined as follows.

$$p(x) = 2x + 2$$
$$q(x) = -x^2 - 2$$

Find the following.

$(p \circ q)(5) =$ ______

$(q \circ p)(5) =$ ______

## Question 5 of 10

A small publishing company is releasing a new book. The production costs will include a one-time fixed cost for editing and an additional cost for each book printed. The total production cost $C$ (in dollars) is given by the function $C = 550 + 16.95N$, where $N$ is the number of books.

The total revenue earned (in dollars) from selling the books is given by the function $R = 34.60N$.

Let $P$ be the profit made (in dollars). Write an equation relating $P$ to $N$. Simplify your answer as much as possible.

## Question 6 of 10

Find the difference quotient $\dfrac{f(x+h)-f(x)}{h}$, where $h \neq 0$, for the function below.

$$f(x) = 8x - 3$$

Simplify your answer as much as possible.

$$\frac{f(x+h)-f(x)}{h} = \underline{\qquad}$$

## Question 7 of 10

Suppose that the functions $f$ and $g$ are defined as follows.

$$f(x) = -5x^2 + 2$$
$$g(x) = \sqrt{x+5}$$

Complete the following.

(a) Find $f \cdot g$. Then give its domain using interval notation.

$(f \cdot g)(x) = \square$

Domain of $f \cdot g$: $\square$

(b) Find $f - g$. Then give its domain using interval notation.

$(f - g)(x) = \square$

Domain of $f - g$: $\square$

(a) Find $f \cdot g$. Then give its domain using interval notation.

$(f \cdot g)(x) =$ ______________________

Domain of $f \cdot g$: ________________

(b) Find $f - g$. Then give its domain using interval notation.

$(f - g)(x) =$ ______________________

Domain of $f - g$: ________________

## Question 8 of 10

For the real-valued functions $g(x)=x^2-5$ and $h(x)=x^2-4$, find the composition $g \circ h$. Also, specify its domain using interval notation.

$(g \circ h)(x)=$

Domain of $g \circ h$:

For the real-valued functions $g(x)=x^2-5$ and $h(x)=x^2-4$, find the composition $g \circ h$. Also, specify its domain using interval notation.

$g \circ h \quad x \quad =$

Domain of $g \circ h$:

## Question 9 of 10

Suppose $H(x)=\sqrt[3]{3x^2+2}$.

Find two functions $f$ and $g$ such that $(f \circ g)(x)=H(x)$.

Neither function can be the identity function.
(There may be more than one correct answer.)

$f(x) =$ ______

$g(x) =$ ______

## Question 10 of 10

The surface area $S(r)$ (in square meters) of a spherical balloon with radius $r$ meters is given by $S(r) = 4\pi r^2$.

The radius $P(t)$ (in meters) after $t$ seconds is given by $P(t) = \frac{4}{3}t$.

Write a formula for the surface area $N(t)$ (in square meters) of the balloon after $t$ seconds.

It is not necessary to simplify.

$N(t) =$

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### Question 1 of 10

$$(g \cdot h)(x) = 2x^3 - 6x^2$$

$$(g + h)(x) = 2x^2 + x - 3$$

$$(g - h)(-1) = -6$$

### Question 2 of 10

(a) $\left(\frac{f}{g}\right)(-1) = -\frac{5}{13}$

(b) Value(s) that are NOT in the domain of $\frac{f}{g}$: $\frac{6}{7}$

### Question 3 of 10

$s(r(-5)) = -8$

Note: The value $s(r(-5))$ is called the composition of the function $s$ with $r$ at $x = -5$.

### Question 4 of 10

$$(p \circ q)(5) = -52$$

$$(q \circ p)(5) = -146$$

### Question 5 of 10

$P = 17.65N - 550$

### Question 6 of 10

$$\frac{f(x+h)-f(x)}{h}=8$$

### Question 7 of 10

(a) Find $f \cdot g$. Then give its domain using interval notation.

$(f \cdot g)(x) = \left(-5x^2+2\right)\sqrt{x+5}$

Domain of $f \cdot g$: $[-5, \infty)$

(b) Find $f-g$. Then give its domain using interval notation.

$(f-g)(x) = -5x^2+2-\sqrt{x+5}$

Domain of $f-g$: $[-5, \infty)$

### Question 8 of 10

$(g \circ h)(x) = x^4 - 8x^2 + 11$

Domain of $g \circ h$: $(-\infty, \infty)$

### Question 9 of 10

$f(x) = \sqrt[3]{x}$

$g(x) = 3x^2 + 2$

### Question 10 of 10

$$N(t) = 4\pi\left(\frac{4}{3}t\right)^2$$